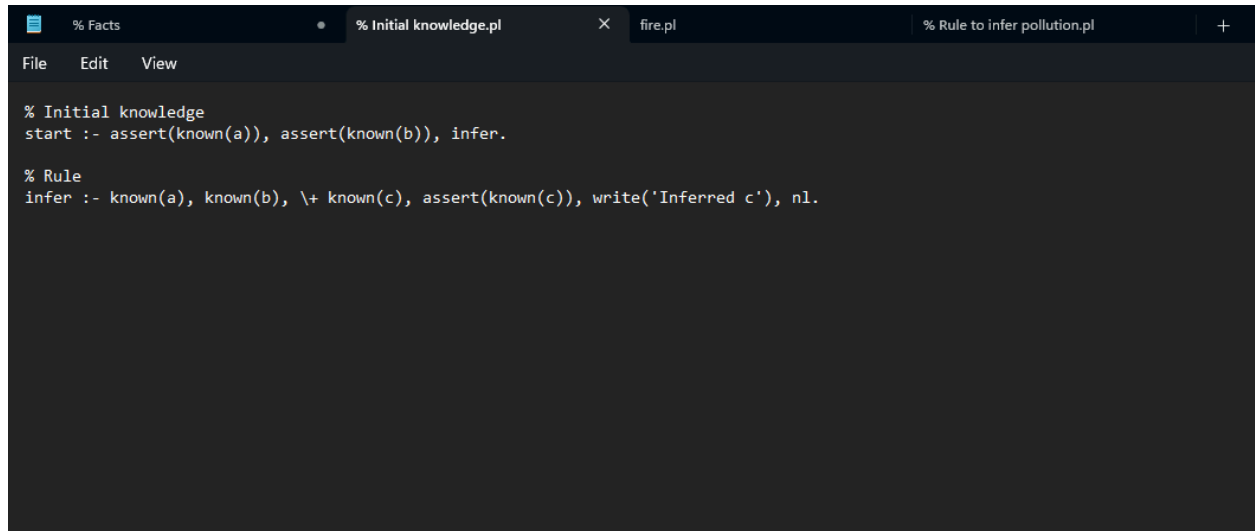


EXERCISE 9 : FORWARD CHAINING

Exercise 1: Simple Inference

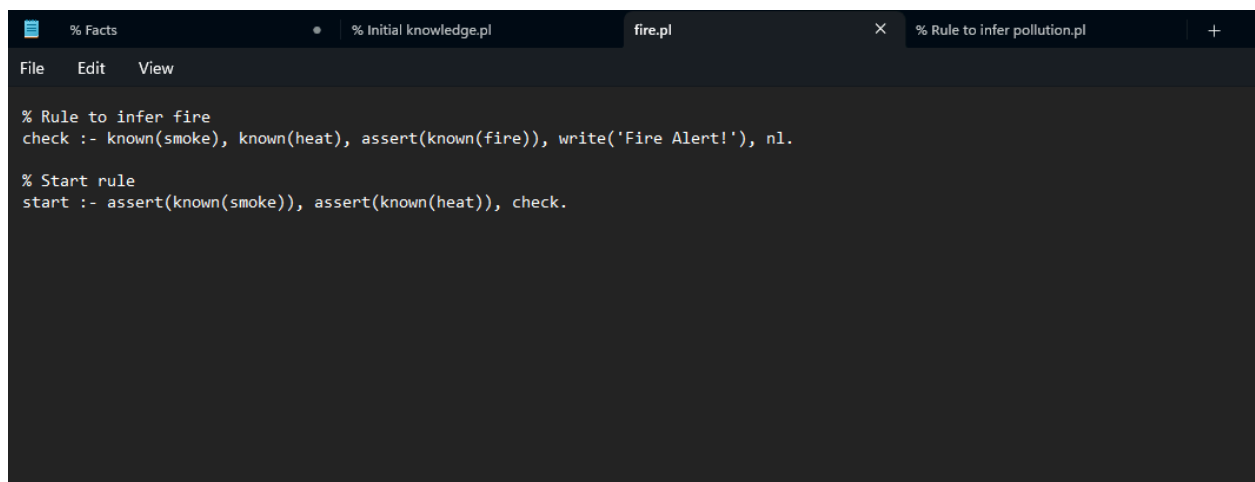


The screenshot shows a Prolog IDE with four tabs: '% Facts', '% Initial knowledge.pl', 'fire.pl', and '% Rule to infer pollution.pl'. The 'fire.pl' tab is active, displaying the following code:

```
% Initial knowledge
start :- assert(known(a)), assert(known(b)), infer.

% Rule
infer :- known(a), known(b), \+ known(c), assert(known(c)), write('Inferred c'), nl.
```

Exercise 2: Fire Alert System



The screenshot shows a Prolog IDE with four tabs: '% Facts', '% Initial knowledge.pl', 'fire.pl', and '% Rule to infer pollution.pl'. The 'fire.pl' tab is active, displaying the following code:

```
% Rule to infer fire
check :- known(smoke), known(heat), assert(known(fire)), write('Fire Alert!'), nl.

% Start rule
start :- assert(known(smoke)), assert(known(heat)), check.
```

Exercise 3: Water Purity Detection

```
% Facts
% Initial knowledge.pl
fire.pl
% Rule to infer pollution.pl

File Edit View

% Rule to infer pollution
infer_pollution :- known(cloudy_water), known(bad_smell), assert(known(polluted)),
write('Water is polluted'), nl.

start :- assert(known(cloudy_water)), assert(known(bad_smell)), infer_pollution.
```

OUTPUT:-

```
SWI-Prolog console
File Settings Tools Debug Help
Welcome to SWI-Prolog (threaded, 64 bits, version 10.0.2)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

6 ?- consult(['c:/Users/HDC0422151/Documents/Prolog/% Initial knowledge.pl']).
true.

6 ?- start.
Inferred c
true.

7 ?- consult(['c:/Users/HDC0422151/Documents/Prolog/fire.pl']).
Warning: c:/users/hdco422151/documents/prolog/fire.pl:5:
Warning: Redefined static procedure start/0
Warning: Previously defined at c:/users/hdco422151/documents/prolog/% initial knowledge.pl:2
true.

8 ?- start.
Fire Alert!
true.

9 ?- consult(['c:/Users/HDC0422151/Documents/Prolog/% Rule to infer pollution.pl']).
Warning: c:/users/hdco422151/documents/prolog/% rule to infer pollution.pl:5:
Warning: Redefined static procedure start/0
Warning: Previously defined at c:/users/hdco422151/documents/prolog/fire.pl:5
true.

10 ?- start.
Water is polluted
true.

11 ?-
| ▲
```